

Comparative Analysis of Image Classification Techniques

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Abstract

This project systematically evaluates and compares prominent image classification techniques—including traditional methods like Bag of Visual Words (BoVW) and its Spatial Pyramid Matching (SPM) extension, spectral feature-based classification, and deep learning architectures such as AlexNet and the Vision Transformer (ViT). Conducted on a local workstation equipped with an NVIDIA RTX 4070 GPU and 32GB of RAM, the study focuses on the Places365 dataset, which was strategically mapped to 4 broad categories for simplified analysis and manageability. Each technique was implemented and evaluated to determine its classification accuracy, computational efficiency, and challenges under these specific hardware constraints. These limitations inherently restricted the feasible number of data samples and the overall model training complexity. Results highlight the critical trade-offs between model complexity, computational requirements, and classification performance across these paradigms. While traditional methods generally offered lower computational overhead, deep learning models consistently achieved higher accuracy, despite facing significant data and memory constraints. This comparative analysis provides practical insights into the applicability and scalability of contemporary image classification methods within common academic or personal computing environments.

1. Introduction

Computer Vision, at its core, seeks to allow machines to interpret and understand visual information within images. A fundamental aspect of this problem is the ability to classify information. For any visual input, the computer's capacity to assign it to a predefined category is often the necessary step to reach complex scene understanding and higher-level analysis. For example, before a machine can successfully understand an image of a dog drinking, it must be capable of reliably identifying the dog as a distinct object in the scene and classifying it.

The rapid evolution of computer vision over the past 20 or so years has led to a diverse set of image classifica-

tion paradigms and unique approaches, ranging from traditional methods rooted in handcrafted features to cutting-edge deep learning architectures. Understanding the relative strengths and weaknesses of these approaches—such as the Bag of Visual Words (BoVW) model [1] and its spatial pyramid matching (SPM) extension [2], spectral feature analysis [3], the foundational AlexNet Convolutional Neural Network (CNN) [4, 5], and the more recent Vision Transformer (ViT) [6]—is crucial for selecting appropriate tools for real-world applications. Each method operates on unique principles, leading to varied tradeoffs of power, robustness, and computational demands. In order to properly guide practitioners in their design choices, comprehensive comparative studies are required.

Virtually all state-of-the-art benchmarks and performances of deep learning initially rely upon immense datasets and highly specialized hardware that may be out of reach for many. Thus, the focus of this project was a comparative analysis of conditions within a constrained environment, allowing for the testing to reflect what real-world limitations each approach may bring. Specifically, this project was run on a local workstation equipped with an NVIDIA RTX 4070 GPU, Intel I7 12700K, and 32GB of RAM, far more realistic for an average user than an entire server rack of computational power. This setup, while capable, imposes practical constraints on the feasible scale of dataset processing and the complexity of models that can be efficiently trained and evaluated. Assessing the performance and resource efficiency of these algorithms under such common academic or personal computing conditions is therefore a key aspect of our motivation.

To permit manageable testing within these limitations, we strategically mapped the large-scale Places365 dataset [7] from 365 total categories down to 4 broad, relatively distinct categories: Indoor Residential, Indoor Public/Commercial, Outdoor Natural, and Outdoor Urban. The goal was to observe the respective capabilities of techniques to distinguish based on true meaning rather than broad template matching. For example, the exact same chair may be present between both residential and commercial settings, yet the orientation, number, and overall presence of this item adds to the context and ultimately grouping. The 4

groupings also allow for a focused analysis of model classification capabilities without the prohibitive memory and computational requirements for 365 fine-grained classes. Therefore, this project seeks to provide a practical comparative analysis, evaluating classification accuracy, training and inference efficiency, and highlighting the specific operational limitations and bottlenecks encountered by each technique within our defined hardware and data scope.

2. Related Works

The problem of image classification has undergone an immense transformation, evolving from simple pattern recognition to highly sophisticated deep learning models capable of near-human performance. Historically, early approaches in computer vision relied almost entirely on handcrafted features and simple matching algorithms [8]. These were improved upon by the later addition of classical machine learning classifiers, permitting higher pattern recognition and the categorization of image content. This paradigm progressively advanced through new techniques, each seeking more effective encoding of visual information into discriminative representations. The introduction of deep neural networks marked a revolutionary shift, enabling a fundamentally new method of learning features from raw image data. These abilities have only been further improved by architectures adapted from natural language processing, such as the Vision Transformer (ViT). These new solutions present an ability to approach the issue of complex classification with general-purpose and highly capable models that continue to push the boundaries of complex classification.

Feature Matching. Prior to the widespread use of machine learning methods, image classification was entirely reliant upon the extraction of handcrafted features and their direct comparison against predefined reference patterns. These techniques focused on identifying key visual characteristics such as edges, textures, and shapes through specific mathematical operations [9]. One such technique, template matching, utilizes computed similarity scores (via sum of squared differences, cross correlation, etc) between a region of interest in an image and a stored template, making it suitable for recognizing rigid objects under controlled conditions [10]. Similarly, edge detection algorithms, including the Sobel operator or more sophisticated Canny edge detector [11], were used to better outline object boundaries for more advanced matching. Histogram analysis, especially color histograms, provided a global descriptor for image content based on pixel values and their distributions [12]. These methods, while foundational, are extremely limited in scope and generalizability. The robustness required for real-world complexities, including variable illumination, scale, rotation, and viewpoint, required the development of improved and invariant feature representation.

Invariant Features. To improve the results of feature matching, the first step was the improvement of the features themselves. Local invariant features arose that were robust to changes in image scaling, rotation, illumination, and small variations in viewpoint, providing more stable and distinctive representations of image content. The Scale-Invariant Feature Transform (SIFT) [13] was the first widespread and successful solution, allowing for the extraction of highly distinctive local keypoints and their associated descriptors that are widely robust to transformations. Efficiency improvements saw the rise of similar techniques such as Speeded Up Robust Features (SURF) [14] and Oriented FAST and Rotated BRIEF (ORB) [15]. The development of such invariant features was critical for advancing object recognition and image matching tasks.

Bag of Visual Words. Classical machine learning approaches such as the support vector machine (SVM) [16] and K-means [17] algorithms combined with invariant features allowed for a massive leap forward from prior matching techniques. Among these prominent traditional approaches, the Bag of Visual Words (BoVW) model emerged as a robust method for image classification, operating similarly to text analysis. The BoVW pipeline operates by detecting local invariant features (such as SIFT, SURF, or ORB) within an image, grouping them into a predefined vocabulary of visual words (learned clustering such as K-means) to build meaning from the features, and then representing each image as a histogram of these visual word frequencies [18]. This method is effective, but it leads to a loss of spatial information as the location of the features is not a factor. To remedy this, the Spatial Pyramid Matching (SPM) extension was developed. SPM operates by partitioning the image into increasingly fine subregions (akin to the Laplacian pyramid) and computing BoVW histograms within each, allowing for critical spatial layout information to be preserved and greatly improving discriminative power [19].

Spectral Features. Aside from local feature grouping, spectral feature analysis offers an alternative approach by transforming image data into a frequency or spectral domain. This method uses properties of signals in these transformed spaces to extract features that describe global image characteristics, textures, or structural regularities not readily apparent in the pixels themselves [3]. Applications range from Fourier Transform-based techniques to more advanced graph-based spectral clustering or embedding approaches. These extracted spectral features are then fed typically into traditional classifiers. The limitation of features chosen and the capabilities of traditional classifiers remain an issue for these approaches, weakening their ability for large-scale classification efforts.

AlexNet and Convolution Neural Networks. Image classification remained largely restricted by classi-

cal machine learning approaches until significant breakthroughs occurred with Convolution Neural Networks (CNNs), largely led by AlexNet [5]. In the 2012 ImageNet Large Scale Visual Recognition Challenge (ILSVRC), the model dominated, outperforming the nearest competitor by 9.8% and showing that deep learning could be extremely effective in practice [20]. AlexNet had several architectural advancements that led to this success, including the use of Rectified Linear Unit (ReLU) activations to accelerate training, Dropout regularization to reduce overfitting, and the pioneering use of Graphics Processing Units (GPUs) for rapid, scalable training. The impact of AlexNet immediately led to research of deeper and more capable CNN architectures, establishing vast libraries of pre-trained CNNs as powerful and versatile feature extractors, as well as the widespread use of transfer learning techniques for many different vision tasks.

Vision Transformers. Beyond CNNs, other deep learning architectures have been extremely effective for visual recognition. Borrowing from the self-attention mechanisms used for Natural Language Processing (NLP), Vision Transformers (ViTs) [21] have shown that purely transformer-based architectures can achieve comparable or better performance on large datasets compared to CNNs, without the biases present with convolution. This approach instead reinterprets images as a series of patches, which are then processed by a transformer encoder to extract global relationships. With the success of ViTs, new areas of research have emerged, ultimately challenging the historical supremacy of CNNs and prompting re-evaluations of architectural choices for fundamental computer vision tasks.

Comparative Analyses and Our Contributions. The individual capabilities and advancements of these diverse image classification techniques have been extensively studied, and numerous comparative analyses exist for them, even for the same dataset as ours. However, such works normally focus on specific areas, such as comparing different CNN architectures [22], or are designed to achieve state-of-the-art results on large benchmarks without consideration for hardware resource limitations [23]. Further, these comparative studies rarely include the wide spectrum from traditional feature-based methods to modern deep learning models, including CNNs as well as ViTs, within a single evaluation. Our project seeks to span this gap by providing a unique comparative analysis with results that are relevant to those with limited hardware resources. We systematically evaluate the performance and real-world operational limitations of Bag of Visual Words (with and without SPM), spectral features, AlexNet (finetuned and feature-extracted), and Vision Transformers. Our testing is done under the constraints of a typical (or slightly above average) consumer-grade workstation (NVIDIA RTX 4070 GPU, Intel i7-12700K CPU, 32GB RAM) and on a strategically

sampled and categorized subset of the Places365 dataset, permitting no more than 100,000 images for total training and testing. This approach offers practical insights into the trade-offs faced by researchers and enthusiasts when using these powerful techniques in resource-limited environments.

3. Methods

3.1. Dataset Preparation

The original, full-size dataset of Places365 is 1.8 million images and nearly 60GB of pixel data alone, separated into 365 unique categories. This scale makes it very difficult to work with in its entirety, thus for experiments a hard limit of 100 thousand images was imposed. Further, the 365 unique categories were hand-mapped into 4 broad categories (Indoor Residential, Indoor Public/Commercial, Outdoor Natural, Outdoor Urban) to make training more manageable and to reduce bias by allowing equal-sized groupings not available in the original 365. The categories were also chosen to highlight the difficulties of semantic distinction, as often the difference between the categories resulted from subtle contextual elements that primitive techniques would struggle to grasp. A split strategy of 80% train, 20% test was used for the images, and these splits were balanced (25 thousand images per group). All images were chosen randomly to avoid bias.

3.2. Traditional Feature-Based Approaches

BoVW and SPM

1. Feature Extraction: CV2 was used to implement SIFT and ORB for local descriptor extraction of the entire 100 thousand image representative set. Due to RAM limitations, this process had to be done in batches consisting of several.pkl files.
2. Visual Vocabulary Creation: The K-Means clustering algorithm was used from `scikit-learn.cluster`, with a K of 1000 visual words. The features extracted were used as the foundational dataset for this stage.
3. Histogram Representation: For each image, the groupings established by K-Means (via nearest neighbor assessment) were used to determine how many features of each "group" were present in that specific image. From these counts, a unique histogram was generated for every image and these histograms were L1 normalized for consistency.
4. Spatial Pyramid Matching (SPM): 2 levels were used for the pyramid due to RAM limitations, resulting in the base level histogram and the 4 quadrant second level histogram for a total of 5 regions. These histograms were concatenated by weighting them (1 for level 0, 1/2 for level 1) and combining them into a single feature vector. This added a layer of spatial information preservation,

yet the limitations did limit the implementation of the technique.

5. Classifier: To utilize GPU speeds the XGBoost library was used for final classification and evaluation with minor hyperparameter tuning (objective: multi:softprob, eval_metric: mlogloss, tree_method: hist, random_state: 42) and GridSearchCV.

Spectral Feature Analysis

1. Feature Extraction: Each image was first resized to a uniform (128, 128) pixels and converted to grayscale. A FFT was then applied to generate the frequency spectrum for each image. The magnitude spectrum was extracted, and a central (32, 32) region around the zero-frequency component was isolated, which was then flattened into a 1024-dimensional feature vector.
2. Classifier: These 1024-D vectors were then used to train an XGBoost classifier identical to the BoVW XGBoost model. Prior to training, the features were normalized by StandardScalar for best results.

3.3. Deep Learning Approaches

AlexNet Fine-Tuning

1. Overview: A pretrained AlexNet model was fine-tuned to measure accuracy.
2. Model Specifics: The AlexNet architecture was initialized from torchvision.models and loaded with weights pretrained on the Places365 dataset. This initial pre-training provided a strong foundation for scene understanding.
3. Architectural Changes: The final classification layer of the pretrained model was replaced with a new nn.Linear layer to output predictions for the 4 broad categories of our dataset rather than the original 365.
4. Training Strategy: The convolutional feature extraction layers were frozen and only the newly added head was trained. This was done to leverage the pretrained power of the model and its learned features.
5. Training Details: We used standard training settings with nn.CrossEntropyLoss() for the loss function, the Adam optimizer with an initial learning rate of 0.0001, the StepLR rate scheduler, 25 epochs, and a batch size of 32 to avoid memory issues with the limited RAM.
6. Data Loading: images and their labels were loaded from a custom HDF5 dataset using PyTorch's built-in methods. This allowed for active image loading from disk rather than loading all images into memory.

AlexNet Feature Extraction

1. Overview: A pretrained AlexNet model was used as a feature extractor, and these features were then used with XGBoost to measure accuracy.
2. Model Specifics: The same AlexNet architecture, pretrained on the Places365 dataset, was used as the feature extractor. Rather than modifying the final layer of the

model, it was modified to output features from its second to last fully-connected layer. This gave a 4096-D feature vector per image, representing high-level information from the dataset.

3. Data Handling: Image data was selectively cached based on preselected indices to manage dataset size; only relevant images were loaded. These numpy arrays were efficiently handled by a TFDSSubsetFeatureDataSet from Pytorch. Features for the training and testing sets were then extracted in batches and saved to HDF5 files for memory-friendly access.
4. Classifier: The extracted 4096-D feature vectors for the training and testing sets were then used to train a separate XGBoost model. This model used the same parameters as seen in the BoVW and Spectral approaches.

Vision Transformer Fine-Tuning

1. Overview: A general purpose ViT trained on the ImageNet-1k dataset was fine-tuned and compared to the AlexNet implementations. This model was not initially trained on the Places365 dataset in order to challenge the extent of the general purpose model.
2. Model Specifics: The ViT base model with a patch size of 16x16 was selected from torchvision.models. It was initialized with pre-trained weights from the ImageNet-1k dataset, providing a strong foundation for general purpose image understanding.
3. Architectural Changes and Training: As with the AlexNet example, the model head was replaced with a new nn.Linear layer configured to output predictions for our 4 categories. This head was the only layer fine-tuned and the rest were frozen. This greatly reduced the number of parameters needed for training, lowering the normally extreme memory and computational demands posed by large models.
4. Training Details: The exact same training settings were used for the ViT as were used in the AlexNet fine-tuning.
5. Data Loading: The same custom HDF5 dataset as the AlexNet experiments was used for efficient memory usage. Preprocessing transforms followed Image-Net ViT standards, including resizing the smallest edge to 256 pixels, taking the center 224x224 crop, and normalizing the images with ImageNet values.

3.4. Computational Environment

All experiments were conducted on a local workstation equipped with an Intel Core i7-12700K CPU, an NVIDIA GeForce RTX 4070 GPU (with 12GB of VRAM), and 32GB of DDR4 RAM. The operating system used was Windows 11. The primary software environment was Python 3.11.9, utilizing key libraries including OpenCV (for image processing and feature extraction), PyTorch (for deep learning model implementation), TensorFlow (for dataset loading from TFDS), scikit-learn (for traditional classifiers and

Table 1. Overall Classification Performance of Different Models on 4-Category Places365 Dataset.

Model Type	Feature Extractor	Classifier	Overall Accuracy (%)	Macro F1-Score (%)
BoVW	SIFT	XGBoost	59.65	59.16
BoVW+SPM	SIFT	XGBoost	60.29	59.77
BoVW	ORB	XGBoost	43.04	42.95
BoVW+SPM	ORB	XGBoost	43.68	43.21
Spectral	FFT	XGBoost	50.86	50.72
AlexNet (Finetuned)	–	CNN Head	85.03	84.97
AlexNet (Feature Extracted)	AlexNet FC (4096-dim)	XGBoost	80.85	80.72
ViT (Finetuned)	–	ViT Head	80.32	80.21

utilities), and XGBoost (for classification). These hardware specifications, particularly the constrained RAM and GPU VRAM, inherently necessitated the use of small batch sizes and limited the total dataset scale to avoid out-of-memory errors and ensure feasible training and inference times for all evaluated methods.

3.5. Evaluation Metrics

The performance of each image classification technique was assessed using a comprehensive set of standard evaluation metrics. These included overall accuracy, as well as precision, recall, and F1-score computed per-class and/or averaged to provide a more nuanced understanding of classification performance. Confusion matrices were generated to visualize the true versus predicted class distributions and identify common misclassifications. Additionally, approximate training were recorded for each method to provide insights into their computational efficiency and practical deployability within the specified hardware environment.

4. Experimental Results and Discussions

4.1. Overall Performance Summary

Experiments were conducted according to the methodology outlined in Section 3. The performance for all evaluated models on the 4-category Places365 dataset were recorded. As presented in Table 1, deep learning approaches exhibited superior overall accuracy and F1-score compared to the traditional feature based methods. The fine-tuned AlexNet model achieved the highest accuracy, followed by the AlexNet feature extraction model. These were both pre-trained on the Places365 dataset, and their results are consistent with expectations regarding transfer learning. The ViT, although it came third, was not pretrained on the Places365 dataset and only its final classification head was fine-tuned. This result underscores the ViT’s inherent power for semantic image classification, even with minimal adaptation to the target domain. The traditional approaches that were purely trained on the 4-category Places365 subset were all consistently 20% or more lower than the deep learning models. This large gap in performance highlights

the superior learning capabilities present in the deep neural networks for scene comprehension.

4.2. Detailed Performance Analysis

BoVW ORB. To best comprehensively understand each model’s capabilities, the following detailed performance analysis will begin by examining the methods in order of increasing overall classification accuracy, starting with the traditional feature-based approaches and finishing with the deep learning models. The lowest performing model in our evaluation was the ORB Bag of Visual Words (BoVW) approach without Spatial Pyramid Matching (SPM), achieving an overall accuracy of only 43%. The confusion matrix for this model, presented in Figure 1, clearly illustrates its primary challenges in distinguishing between semantically similar categories. Specifically, categories with nuanced contextual differences, such as ‘Indoor Residential’ and ‘Indoor Public/Commercial’ scenes, were frequently confused. A similar pattern of confusion was observed between ‘Out-

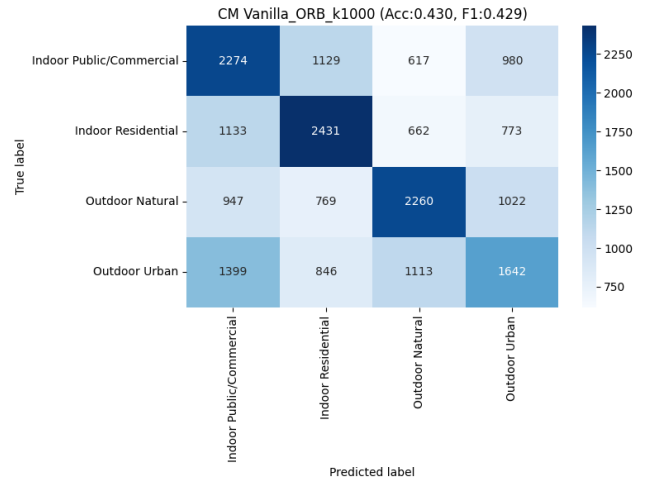


Figure 1. Confusion Matrix for BoVW (ORB, Non-SPM) on the 4-Category Places365 Test Set (Accuracy: 43.04%). Categories: 0: Indoor Public/Commercial, 1: Indoor Residential 2: Outdoor Natural, 3: Outdoor Urban.

door Natural' and 'Outdoor Urban' environments. One of the most significant systematic errors involved the frequent misclassifications of 'Outdoor Urban' scenes as 'Indoor Public/Commercial' (or vice-versa), indicating a fundamental struggle to capture the distinctive global characteristics of these environments. This performance suggests that ORB descriptors, while computationally efficient, may not provide sufficiently discriminative or descriptive local features for meaningful classification across these broad scene categories. Furthermore, the inclusion of SPM with ORB features did not yield a significant performance improvement. This outcome is likely attributed to the hardware limitations on pyramid levels, which restricted the granularity of spatial information that could be effectively incorporated, thus preventing robust conclusions from being drawn regarding the benefit of spatial pooling for ORB in this context.

Spectral Features. The global details retained by the spectral features proved superior to ORB features, achieving an overall accuracy of 50.86%. The confusion matrix for this model, Figure 2, reveals that even with these global details there were inherent limitations that persisted. Consistent with the ORB analysis, the model still frequently confused categories with subtle contextual differences, such as 'Indoor Residential' with 'Indoor Public/Commercial' scenes, and 'Outdoor Natural' with 'Outdoor Urban' environments. One of the most significant systematic errors involved the persistent misclassifications of 'Outdoor Urban' scenes as 'Indoor Public/Commercial' (or vice-versa), mirroring the challenge observed with ORB. This outcome suggests that while spectral features excel at identifying

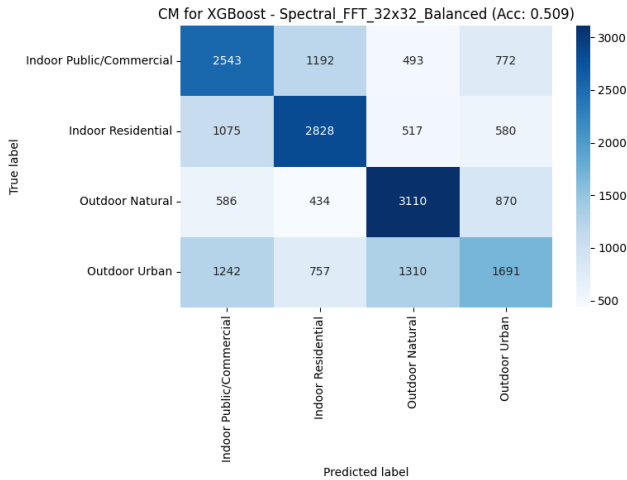


Figure 2. Confusion Matrix for BoVW (ORB, Non-SPM) on the 4-Category Places365 Test Set (Accuracy: 50.86%). Categories: 0: Indoor Public/Commercial, 1: Indoor Residential 2: Outdoor Natural, 3: Outdoor Urban.

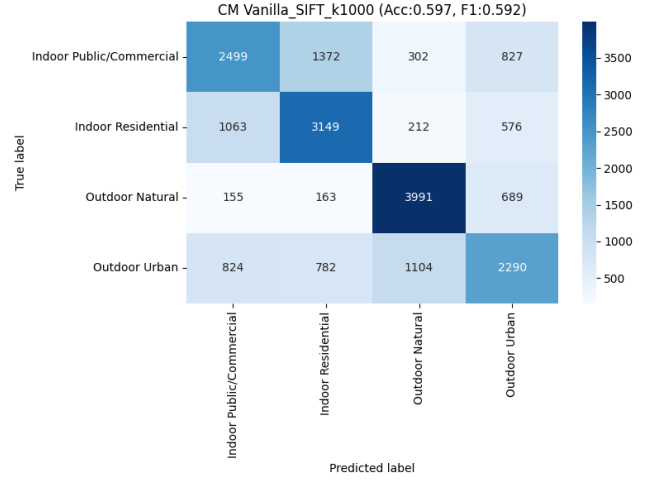


Figure 3. Confusion Matrix for BoVW (SIFT, Non-SPM) on the 4-Category Places365 Test Set (Accuracy: 59.65%). Categories: 0: Indoor Public/Commercial, 1: Indoor Residential 2: Outdoor Natural, 3: Outdoor Urban.

dominant textural patterns and structural regularities (e.g., strong horizontal/vertical lines, repeating elements common in both urban and indoor architecture), they may not adequately encode the higher-level semantic cues or specific spatial layouts essential for nuanced distinctions between these contexts. The shared failure mode with ORB in distinguishing urban from indoor commercial environments underscores a fundamental challenge for purely feature-based approaches in tasks requiring deeper semantic understanding.

BoVW SIFT. Transitioning back to local feature approaches within the BoVW pipeline, SIFT descriptors yielded a substantial performance improvement over ORB, achieving an overall accuracy of 59.65%, a 17% increase in accuracy. Furthermore, SIFT-based BoVW surpassed the performance of the global spectral features, indicating that despite their local nature, SIFT descriptors captured more discriminative scene information. The confusion matrix for this model (Figure 3) illustrates that while overall accuracy improved, the inherent limitations of hand-crafted features remained evident. Persistent confusion was observed between semantically similar categories, specifically 'Indoor Residential' and 'Indoor Public/Commercial' scenes, and 'Outdoor Natural' and 'Outdoor Urban' environments. However, a significant breakthrough was observed in differentiating 'Outdoor Urban' scenes from 'Indoor Public/Commercial' environments, a pair frequently misclassified by both ORB and spectral methods. This suggests SIFT's robustness to scale, rotation, and illumination, combined with its denser and more distinctive local feature representation, better enabled it to capture the unique struc-

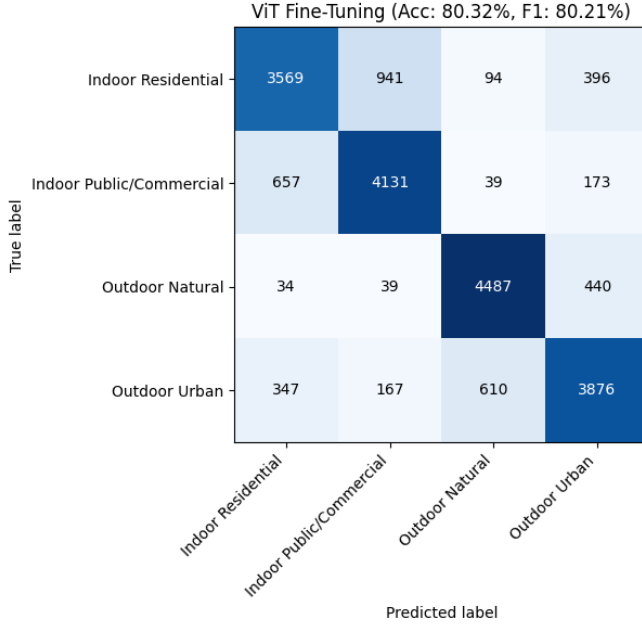


Figure 4. Confusion Matrix for ViT (Fine-Tuning) on the 4-Category Places365 Test Set (Accuracy: 80.32%). Categories: 0: Indoor Public/Commercial, 1: Indoor Residential 2: Outdoor Natural, 3: Outdoor Urban.

tural and textural elements that distinguish these two categories. Consistent with prior observations, the application of Spatial Pyramid Matching (SPM) to SIFT features similarly yielded no substantial performance gain, an outcome still attributed to the hardware limitations on pyramid levels that restricted the granularity of spatial information effectively incorporated.

ViT. While a state-of-the-art model designed for global context, the Vision Transformer (ViT) architecture achieved an overall classification accuracy of 80.32%. This performance, while a substantial improvement over earlier techniques, was slightly surpassed by the standard AlexNet model and more notably by the fine-tuned AlexNet variant (which reached 85.03%). This disparity is primarily attributable to fundamental differences in their pretraining datasets. The evaluated ViT model was pre-trained on the object-centric ImageNet-1k dataset, unlike AlexNet’s pre-training directly on Places365. This domain mismatch likely harmed ViT’s initial ability to fully capture the high-level contextual and global scene features critical for our broad category classification task without further extensive fine-tuning on Places365. However, even with this pre-training disadvantage, ViT consistently displayed strong semantic and learning capabilities. Its confusion matrix (Figure 4) reveals that, while some marginal confusion persisted between ‘Outdoor Urban’ and ‘Outdoor Natural’ environments, and ‘Indoor Residential’ and ‘Indoor Pub-

lic/Commercial’ scenes, these misclassifications were significantly reduced compared to the best-performing BoVW approach. This outcome underscores ViT’s remarkable generalization capabilities, while simultaneously highlighting the critical importance of domain-aligned pretraining for maximizing performance in specific classification tasks, particularly when computational resources for extensive fine-tuning are limited.

AlexNet. Transitioning to deep learning paradigms, the AlexNet architecture, leveraging features extracted from its pre-trained convolutional layers, demonstrated a substantial leap in classification accuracy, achieving an overall performance of 80.85%. Notably, fine-tuning the classification head of this pre-trained network further enhanced performance to 85.03%, a result that surpassed even the state-of-the-art ViT model when not specifically domain-trained on Places365. The AlexNet (Feature Extraction) variant exhibited confusion patterns similar to those observed with the ViT model, particularly in distinguishing ‘Outdoor Urban’ from ‘Outdoor Natural’ environments, and ‘Indoor Residential’ from ‘Indoor Public/Commercial’ scenes. However, AlexNet, especially the fine-tuned version, provided a significant mitigation of these confusions, as illustrated by its confusion matrix in Figure 5. While minor confusion persisted in the categories mentioned, the fine-tuning process allowed for further reduction in these classification errors. Furthermore, the fine-tuned AlexNet model achieved more than double the accuracy of the baseline ORB-BoVW implementation, showcasing the profound impact of deep

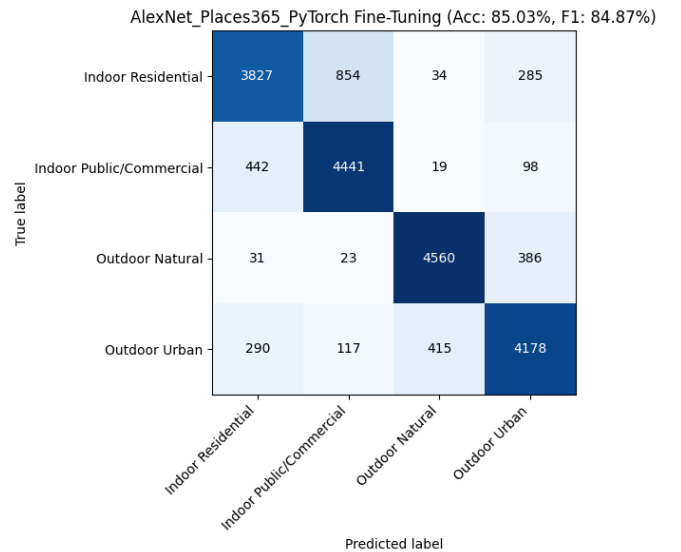


Figure 5. Confusion Matrix for AlexNet (Fine-Tuning) on the 4-Category Places365 Test Set (Accuracy: 85.03%). Categories: 0: Indoor Public/Commercial, 1: Indoor Residential 2: Outdoor Natural, 3: Outdoor Urban.

learning and domain-aligned pre-training on complex scene classification tasks.

4.3. Computational Efficiency

The computational demands and training durations varied significantly across the evaluated approaches for the 100,000-image dataset. An initial data setup phase (including dataset download and splitting), common to all methods, required approximately 120 minutes (though this is highly dependent on internet and storage speeds).

The spectral feature extraction and subsequent classifier training proved to be the most computationally efficient, completing in approximately 20 minutes. This rapid processing is attributed to the inherent speed of Fast Fourier Transforms (FFTs) for feature generation. For the Bag of Visual Words (BoVW) pipelines, overall execution times ranged from 6 to 9 hours. This period primarily encompasses the computationally intensive processes of local feature detection and description (ORB, SIFT) across all images, visual vocabulary creation, and the final XGBoost classifier training. Non-SPM ORB was consistently the fastest among these, while SPM SIFT was the longest.

In contrast, the deep learning approaches (AlexNet and ViT) demonstrated a different profile of efficiency. While benefiting immensely from pre-trained weights, their training phases were still substantial. The AlexNet feature extraction, which involves a single pass through the pre-trained network to generate features for the entire dataset, took about 45 minutes. The fine-tuning of both AlexNet and ViT classification heads, which involves iterative back-propagation for many parameters, each required nearly 2 hours.

It is important to note that for all approaches, consistently conservative hyperparameter choices were made, prioritizing minimized training time over maximizing potential accuracy. This conscious trade-off underscores the practical considerations of running such experiments within the constraints of a local workstation.

5. Conclusion

This project systematically evaluates diverse image classification paradigms—including traditional Bag of Visual Words with Spatial Pyramid Matching, spectral feature-based classification, and modern deep learning architectures like AlexNet and Vision Transformer—on a 4-category Places365 subset under common workstation constraints. Our findings affirm the superior performance of deep learning architectures, particularly when leveraging transfer learning, over traditional methods. Crucially, this study illuminates the practical performance-resource trade-offs faced by researchers and practitioners on moderately equipped systems, demonstrating that competitive classification accu-

racy with deep learning is attainable even within such confines.

6. Contributions and Acknowledgments

Christopher Kaelin was responsible for all code gathering, debugging, experimentation, and analysis. This report was drafted and polished by him. He also was responsible for the associated YouTube video.

Assistance with language and structural refinement, was provided by Google Gemini 2.5 Flash and Grammarly. The intellectual content and any errors remain the author's own.

References

- [1] J. Yang, Y.-G. Jiang, A. G. Hauptmann, and C.-W. Ngo, "Evaluating bag-of-visual-words representations in scene classification," in *Proceedings of the international workshop on Workshop on multimedia information retrieval*, 2007, pp. 197–206. [1](#)
- [2] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proceedings of the 18th SIGSPATIAL international conference on advances in geographic information systems*, 2010, pp. 270–279. [1](#)
- [3] Y. Ozaki, S. Morita, and Y. Du, "Spectral analysis," *Near-infrared spectroscopy in food science and technology*, pp. 47–72, 2007. [1](#), [2](#)
- [4] M. Z. Alom, T. M. Taha, C. Yakopcic, S. Westberg, P. Sidike, M. S. Nasrin, B. C. Van Esesn, A. A. S. Awwal, and V. K. Asari, "The history began from alexnet: A comprehensive survey on deep learning approaches," *arXiv preprint arXiv:1803.01164*, 2018. [1](#)
- [5] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Alexnet - imagenet classification with deep convolutional neural networks." [1](#), [3](#)
- [6] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020. [1](#)
- [7] B. Zhou, A. Lapedriza, A. Khosla, A. Oliva, and A. Torralba, "Places: A 10 million image database for scene recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 40, no. 6, pp. 1452–1464, 2017. [1](#)
- [8] D. M. Mount, N. S. Netanyahu, and J. Le Moigne, "Efficient algorithms for robust feature matching," *Pattern recognition*, vol. 32, no. 1, pp. 17–38, 1999. [2](#)
- [9] R. Szeliski, *Computer vision: algorithms and applications*. Springer Nature, 2022. [2](#)
- [10] R. Brunelli, *Template matching techniques in computer vision: theory and practice*. John Wiley & Sons, 2009. [2](#)
- [11] J. Canny, "A computational approach to edge detection," *IEEE Transactions on pattern analysis and machine intelligence*, no. 6, pp. 679–698, 1986. [2](#)
- [12] M. J. Swain and D. H. Ballard, "Indexing via color histograms," in *Active perception and robot vision*. Springer, 1992, pp. 261–273. [2](#)

- [13] G. Lowe, "Sift-the scale invariant feature transform," *Int. J.*, vol. 2, no. 91-110, p. 2, 2004. 2
- [14] H. Bay, T. Tuytelaars, and L. Van Gool, "Surf: Speeded up robust features," in *Computer Vision—ECCV 2006: 9th European Conference on Computer Vision, Graz, Austria, May 7-13, 2006. Proceedings, Part I 9*. Springer, 2006, pp. 404–417. 2
- [15] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "Orb: An efficient alternative to sift or surf," in *2011 International conference on computer vision*. Ieee, 2011, pp. 2564–2571. 2
- [16] T. Joachims, "Svmlight: Support vector machine," *SVM-Light Support Vector Machine <http://svmlight.joachims.org/>*, University of Dortmund, vol. 19, no. 4, p. 25, 1999. 2
- [17] D. Steinley, "K-means clustering: a half-century synthesis," *British Journal of Mathematical and Statistical Psychology*, vol. 59, no. 1, pp. 1–34, 2006. 2
- [18] Sivic and Zisserman, "Video google: A text retrieval approach to object matching in videos," in *Proceedings ninth IEEE international conference on computer vision*. IEEE, 2003, pp. 1470–1477. 2
- [19] S. Lazebnik, C. Schmid, and J. Ponce, "Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories," in *2006 IEEE computer society conference on computer vision and pattern recognition (CVPR'06)*, vol. 2. IEEE, 2006, pp. 2169–2178. 2
- [20] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012. 3
- [21] I. O. Tolstikhin, N. Houlsby, A. Kolesnikov, L. Beyer, X. Zhai, T. Unterthiner, J. Yung, A. Steiner, D. Keysers, J. Uszkoreit *et al.*, "Mlp-mixer: An all-mlp architecture for vision," *Advances in neural information processing systems*, vol. 34, pp. 24 261–24 272, 2021. 3
- [22] M. Tan and Q. V. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International Conference on Machine Learning*. PMLR, 2019, pp. 6105–6114. 3
- [23] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 2009, pp. 248–255. 3